

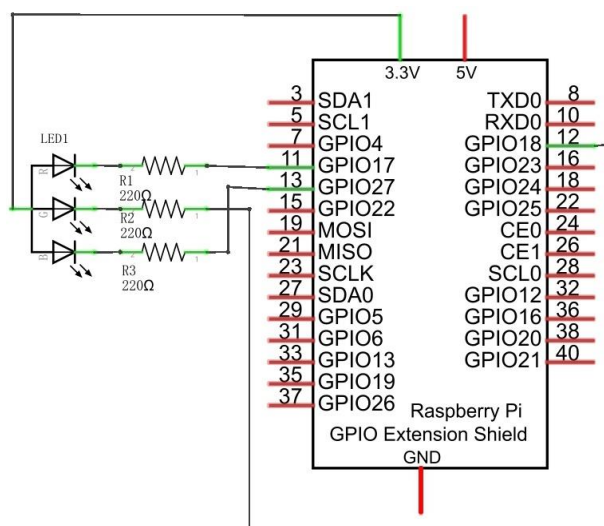
## Project 5.1 Multicolored LED

### Component List

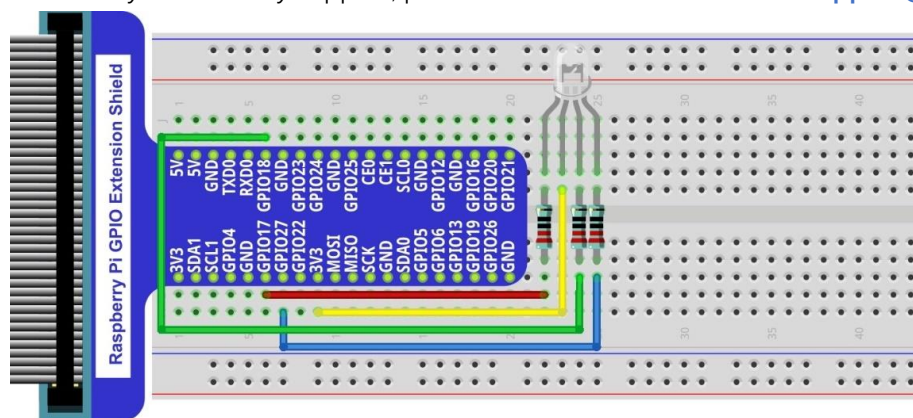
|                                                                                                  |                                                                                                 |                                                                                                         |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Raspberry Pi (with 40 GPIO) x1<br>GPIO Extension Board & Wire x1<br>Breadboard x1                | RGB LED x1<br> | Resistor 220Ω x3<br> |
| Jumper Wire<br> |                                                                                                 |                                                                                                         |

### Circuit

Schematic diagram



Hardware connection. If you need any support, please feel free to contact us via: [support@freenove.com](mailto:support@freenove.com)

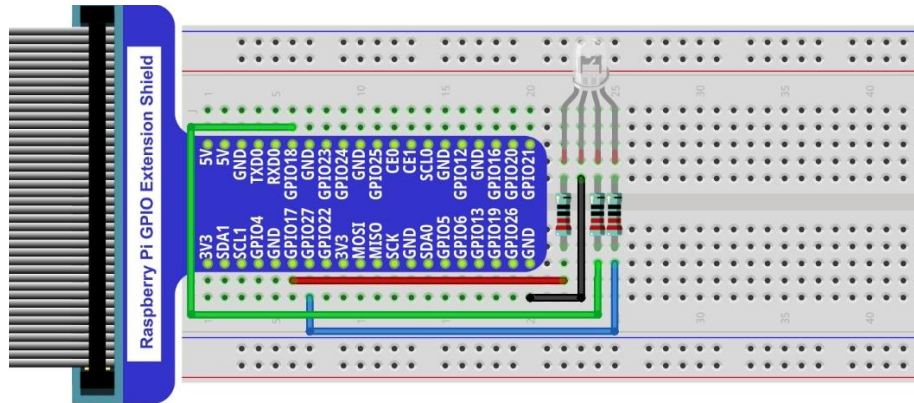


Video: <https://youtu.be/tbnX2AsX2y4>

In this kit, the RGB led is **Common anode**. The **voltage difference** between LED will make it work. There is no visible GND. The GPIO ports can also receive current while in output mode.

If circuit above doesn't work, the RGB LED may be common cathode. Please try following wiring.

There is no need to modify code for random color.



## Code

We need to use RGBLED class to control RGBLED. The parameters for setting the RGBLED as common cathode or common anode are provided in the RGBLED class. You can set it according to the type of your RGB LED, and the default setting in our example code is based on common anode.

### Python Code 5.1.1 ColorfulLED

First, observe the project result, and then learn about the code in detail.

If you have any concerns, please contact us via: [support@freenove.com](mailto:support@freenove.com)

1. Use cd command to enter 05.1.1\_ColorfulLED directory of Python code.

```
cd ~/Freenove_Kit/Code/Python_GPIOWireless/05.1.1_ColorfulLED
```

2. Use python command to execute python code "ColorfulLED.py".

```
python ColorfulLED.py
```

After the program is executed, you will see that the RGB LED randomly lights up different colors.

The following is the program code:

```
1 from gpiozero import RGBLED
2 import time
3 import random
4
5 #led = RGBLED(red="J8:11", green="J8:12", blue="J8:13", active_high=False) # define the pins
  for R:11,G:12,B:13
6 led = RGBLED(red=17, green=18, blue=27, active_high=False) # define the pins for
  R:GPIO17,G:GPIO18,B:GPIO27
7 # If your RGBLED is a common cathode LED, set active_high to True
8
9 def setColor(r_val, g_val, b_val):      # change duty cycle for three pins to r_val, g_val, b_val
10     led.red=r_val/100                  # change pwmRed duty cycle to r_val
11     led.green = g_val/100              # change pwmRed duty cycle to r_val
12     led.blue = b_val/100               # change pwmRed duty cycle to r_val
13
14 def loop():
```

```

15     while True :
16         r=random.randint(0,100) #get a random in (0,100)
17         g=random.randint(0,100)
18         b=random.randint(0,100)
19         setColor(r,g,b)          #set random as a duty cycle value
20         print ('r=%d, g=%d, b=%d ' %(r ,g, b))
21         time.sleep(1)
22
23 def destroy():
24     led.close()
25
26 if __name__ == '__main__':      # Program entrance
27     print ('Program is starting ... ')
28     try:
29         loop()
30     except KeyboardInterrupt: # Press ctrl-c to end the program.
31         destroy()
32     print("Ending program")

```

Import the RGBLED class that controls RGBLED from the gpiozero library.

```
from gpiozero import RGBLED
```

Create the RGBLED class for controlling the RGBLED.

```
led = RGBLED(red=17, green=18, blue=27, active_high=False) # define the pins for
R:GPIO17,G:GPIO18,B:GPIO27
```

In the previous chapter, we learned how to make a pin output PWM using the Python language. In this project, we output to three pins via PWM. In the "while" loop of the "loop" function, we first generate three random numbers and then assign these three random numbers to the PWM values of the three pins, which will make the RGB LED randomly produce multiple colors.

```

def loop():
    while True :
        r=random.randint(0,100) #get a random in (0,100)
        g=random.randint(0,100)
        b=random.randint(0,100)
        setColor(r,g,b)          #set random as a duty cycle value
        print ('r=%d, g=%d, b=%d ' %(r ,g, b))
        time.sleep(1)

```

For more information about the methods used by the RGBLED class in the GPIO Zero library, please refer to:

[https://gpiozero.readthedocs.io/en/stable/api\\_output.html#rgbled](https://gpiozero.readthedocs.io/en/stable/api_output.html#rgbled)